Exploring AI Adoption in Regional HEIs: Faculty Readiness and Use in Azerbaijan

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Abstract. In modern life, the artificial intelligence (AI) starting to play an increasingly significant role, its subsequent integration into higher education institutions (HEIs) has become an important component of digital transformation. While AI demonstrates the huge potential to improve the effectiveness in both teaching and administrative operations, its adoption has not been uniform across all institutions. In regions with lower levels of digital literacy and underdeveloped infrastructure-particularly in remote areas- AI is not yet being used effectively or consistently. These disparities make it essential to examine how educators are using AI and what challenges they face, in order to inform more inclusive and effective digital transition policies. This study investigates the awareness, usage patterns, and challenges related to AI tools among academic staff in regional HEIs in Azerbaijan. The survey involved 492 respondents (educators) from 13 institutions of various ages and experience levels and was conducted via the Qualtrix platform. The survey assessed digital readiness, familiarity with AI tools (e.g., ChatGPT), frequency of use, influencing factors, perceived benefits, and challenges encountered. Descriptive analysis revealed that approximately 40% of respondents use AI frequently or daily, while 17.9% are not familiar with AI tools. A statistically significant relationship was found between age and AI usage, with younger academic staff showing higher familiarity and more frequent use. This research is directed to uncover the present Aİ adoption situation in Azerbaijan regional universities and shows key areas for targeted interventions. The result can be considered in the development of digital strategy for higher education institutions.

Keywords: Artificial intelligence · Higher education institutions · digital transformation · educators' digital competence

1 Introduction

The adoption of Artificial İntelligence (Aİ) in higher education sector is growingly reshaping traditional paradigms of knowledge delivery, learning management and further graduate employability. As AI technologies are constantly enhanced, HEIs need to move beyond considerations of technical adoptability to critical approach the pedagogical compliance ethical implications, and infrastructural requirements of further seamless AI adoption. Authors at [1] mentioned the proliferation of generative tools as ChatGT

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has accelerated this shift, pushing institutions to redesign educational practices and governance structures. AI-driven platforms introduce significant opportunities and complex challenges and play pivotal roles not only in administrative tasks, but also in curriculum development, as well as student assessment, and development of personalized learning pathways.

A major challenge arising from the prevalent use of generative AI tools is the issue as trust and authenticity in academic works. As [2] emphasizes, AI systems produce human-like content, which becomes crucial to establish clear boundaries between machine-generated and human intellectual output. Without such differentiation the principals of academic integrity risk being violated. Dignum at [3] also states that transparency and responsibility must predicate the use of AI in education to prevent biases, misinformation and ethical violations. Subsequently, HEIs must develop robust frameworks that ensue the alignment od AI technologies with core academic principles, that emphasize critical thinking, originality and social responsibility.

The adoption of AI being a widespread technological trend is also a transformative force demanding an interdisciplinary and value-driven institutional reaction. In the context of rapid digitalization investments in faculty training, ethical governance frameworks, and redundant technical infrastructure are important to preserve the human-centric values of higher education.

1.1 Problem Statement

Several authors highlighted the application of AI tools for various purposes in HEIs processes such as grading, intelligent tutoring, and data-supported feedback mechanisms [4, 5]. These advantages at the same time bring out concerns on academic integrity, algorithmic bias, and disruption of critical thinking skills [6, 7]. For this particular reason European Commission (2019) has recognized the requirements for AI tools in education processes to be lawful, ethical and socially sustainable [8].

In regional universities mostly being under-resourced and observing infrastructural restrictions, as well as low digital literacy face even more pronounced challenges in effective and equitable AI adoption [9, 10]. In this context, HEIs awareness and preparedness, related governing, controlling and monitoring policies and support mechanisms being in place, determine whether AI will serve educational innovation or enforce existing inequalities. Furthermore, the lack of proper solutions on AI governance, academic freedom, and human agency in decision-making [2, 8] hinder the AI adoption process.

HEIs risk falling behind in technological progress without context-sensitive and ethical grounded policies. In other scenario universities integrate AI that compromise their educational missions lacking related frameworks in place.

1.2 Research Objectives

This research aims to contribute to the increasing literature on AI adoption in HEIs, particularly focusing on the regional university faculty readiness in under-resourced context. Particularly, the study aims to:

- Assess faculty readiness, attitudes, and observed obstacles regarding AI tools.

- Investigate how institutional factors, educator experience, and demographic variables effects AI adoption
- Provide actionable insights that inform policy development and strategic planning for responsible AI integration in regional universities.

Through this localized focus, the study aspires to enrich the global discourse with perspectives that move beyond technologically advanced centers and better represent diverse educational realities.

1.3 Research Questions

To guide the investigation, the following research questions were formulated:

1. To what extent are faculty members in regional universities aware of AI tools and their educational applications?
2. How are AI technologies currently being utilized by faculty members in their teaching and research activities?
3. What challenges and barriers inhibit the effective adoption of AI at the institutional level?
4. Do demographic variables such as age, gender, or academic affiliation influence perceptions of and readiness for AI integration?

1.4 Significance of the Study

This study holds both academic and practical significance. Academically, it contributes to the limited body of empirical research focusing on AI adoption in under-resourced higher education contexts, thereby filling an important gap in the existing literature that often privileges technologically advanced institutions. Practically, the findings will offer evidence-based insights for university administrators, policymakers, and educators, helping them design targeted interventions that enhance digital readiness, promote ethical AI use, and mitigate structural barriers to adoption.

Moreover, by emphasizing a context-sensitive approach, the study advocates for an inclusive educational technology discourse that recognizes the diverse challenges and needs of regional universities. It emphasizes the importance of investing on educators' development and ethical governance frameworks, as well as digital infrastructure. The findings from the study can develop policies for AI adoption that can provide equitable, social responsible, and human-centric learning outputs within different HEIs context.

2 Literature Review

In the past decade the academic debate over AI usage in higher education The academic discourse surrounding Artificial Intelligence (AI) in higher education has gained importance. This tendency highly correlates with introduction of generative AI tools as ChatGPT. Researchers have tackled the topic from variety of context, such as pedagogical innovation, ethical responsibility, faculty preparedness, and learner-instructor relations. Though major body of literature accepts the transformative contribution of AI, at the time within unregulated or uncritical terms it creates various risks and restrictions.

An increasing number of researches concentrated on the dual specifications of generative AI tools. From one side, they are prioritized for improving education efficiency automated assessments, content creation, and personalized feedback. On the other side, academic integrity, usage of unverified content, and risks around weakening of critical thinking skills raised concerns. According Michel-Villarreal et al. (2023) ChatGPT and similar tools can jeopardize tradition grading and authorship models. In this context higher education is not prepared to address this challenge. In their study they gave ChatGPT question about its role in education. After exploring responces they came to conclusion that for effective usage of AI clear institutional policies and guidance are urgently required.

In their comprehensive review [1] through systematic analyses of peer-reviewed studies investigated the implications of CharGPT in higher education. They distinguished the most popular uses of technology, such as academic writing and tutoring support, automated assessment, but they also highlighted persisting ethical and regulatory concerns. A primary contribution of their research is a set of practical guidelines for responsible usage that prioritize honesty, critical interaction, and educator supervision. Additionally, they claimed that to prohibit these tools are ineffective and pointless, and instead encourage flexibible institutional policies that embrace AI's presence in academic settings.

The ethical framework that should guide the usage of AI is another essential aspect. European Union's guidelines for "Trustworthy AI" guidelines and its relevance to education are brought up by [11]. They suggest that related trainings on responsible AI practices for both students and educators should be provided to enforce ethical use of AI beyond regulatory compliance. Their research reveals the concern that although HEIs are interested in AI ethics, its actual implementation in higher education curriculum remains mostly inconsistent and partial. The researchers consistently emphasize the necessity of integration of ethical requirements across educational processes to provide graduates with not only technological literacy, but also social responsibility while using AI.

The AI influence on educational processes especially relationships between students and instructors has been the focus area for many researchers as well. Seo et al. (2021) investigate the effect of using AI tools on communication, support, and engagement in online learning context. They came into conclusion that Ai on the one hand, can improve efficiency and availability, on the other hand, it can hinder the human participation in teaching and learning. As an example, they mentioned reduced interaction and sense of human elements resulted from usage of AI tutors or automated assessment tools. The authors highlight the importance of maintaining a "human-in-the-loop" design philosophy to ensure that AI tools supplement rather than replace meaningful engagement between educators and learners.

A broader review of the state of AI in higher education is offered by [5], who conducted a systematic analysis of 138 peer-reviewed articles published between 2016 and 2022. Their findings indicate a rapid increase in AI-related publications, especially in the years following the COVID-19 pandemic. The study identifies five primary uses of AI: assessment and evaluation, predictive analytics, AI assistants, intelligent tutoring

systems, and student learning management. Notably, 72% of the studies focused on students, with only 17% addressing instructors. This imbalance suggests that while student-centered applications are flourishing, there is less emphasis on faculty development and institutional capacity-building- a critical gap in the literature.

What emerges across these studies is a consensus on the urgent need for structured institutional responses to AI integration through ethical frameworks, pedagogical guidelines, or faculty training programs. The literature points to a collective recognition that, AI is not a temporary trend but a permanent feature of the educational landscape. However, there remains a significant disparity in how institutions—particularly those in regional or under-resourced contexts are equipped to respond to this shift. The lack of uniform policy, infrastructure, and expertise may hinder equitable access to AI's benefits, potentially exacerbating existing inequalities in higher education.

In summary, while the literature presents a rich and evolving discussion on AI in higher education, it also highlights areas that require further exploration. These include localized studies on faculty readiness, the long-term impact on student learning behaviors, and the development of scalable models for ethical AI integration. By addressing these gaps, future research can contribute to a more inclusive and sustainable digital transformation of education.

While existing scholarship provides valuable insights into the pedagogical, ethical, and institutional dimensions of AI in higher education, much of it remains concentrated on technologically advanced or well-resourced institutions. There is comparatively limited empirical research examining how faculty members in regional or underfunded universities perceive and engage with AI tools. This study seeks to address that gap by investigating digital readiness, usage patterns, and perceived barriers among academic staff in a regional context. Through a mixed-methods approach, it aims to uncover the localized challenges and opportunities associated with AI integration and inform more equitable strategies for institutional support.

This chapter reviewed the evolving academic discourse on the role of Artificial Intelligence in higher education, highlighting both its transformative potential and the challenges it presents. The literature reveals a dual perspective: while AI tools like ChatGPT offer significant benefits in terms of efficiency, assessment, and personalized learning, they also raise serious concerns regarding ethics, critical thinking, and institutional readiness. Key themes such as the need for ethical guidelines, maintaining human-centered pedagogy, and bridging infrastructural gaps were consistently emphasized. Moreover, the review identified a notable research gap concerning AI integration in regional or underfunded higher education institutions. Addressing this gap, the present study aims to explore localized faculty experiences and readiness, thereby contributing to a more inclusive and context-sensitive understanding of AI's role in education.

3 Methodology

3.1 Research Design

This study employed a quantitative survey-based research design to investigate university faculty members' perceptions, readiness, and application of Artificial Intelligence (AI) technologies in a regional higher education context. A cross-sectional approach was

adopted to capture a snapshot of faculty attitudes and digital preparedness at a particular point in time, following established guidelines for educational survey research [12, 13].

The research aimed to answer the following guiding questions:

1. To what extent are faculty members in regional universities aware of AI tools and their educational applications?
2. How are AI technologies currently being used by faculty in teaching and research?
3. What challenges and barriers inhibit effective AI adoption at the institutional level?
4. Do variables such as age, gender, or academic affiliation affect perceptions and readiness?

3.2 Participants and Sampling

The survey was administered to academic staff across various faculties of regional public universities in Azerbaijan. A total of 730 valid responses were collected, representing a wide range of disciplines including humanities, social sciences, STEM, education, and management. Participants included lecturers, assistant professors, associate professors, and full professors.

A stratified purposive sampling strategy was adopted to ensure diversity in academic ranks and departments. Respondent demographics included age, gender, institutional affiliation, and years of teaching experience, which were later used for subgroup comparisons.

3.3 Instrumentation

Data were collected using a structured online questionnaire developed by the researchers. The questionnaire was validated by experts in educational technology and consisted of four key sections:

1. Demographic Information – age, gender, university, academic rank, years of experience.
2. AI Awareness and Usage – familiarity with AI platforms such as ChatGPT, frequency and purpose of use.
3. Attitudes Toward AI – measured on a 5-point Likert scale (strongly disagree to strongly agree) covering perceived usefulness, concerns, and trust in AI tools.
4. Perceived Challenges – including infrastructure issues, lack of training, and student misuse.

The internal consistency of the instrument was assessed using Cronbach's alpha, which yielded a reliability coefficient of 0.86, indicating strong internal reliability.

3.4 Data Collection Procedure

The survey was distributed via institutional emails, internal platforms, and academic networks over a four-week period. Participants were informed about the purpose of the study, assured of anonymity, and asked to provide informed consent before responding. The study was conducted in accordance with ethical standards for educational research, and informed consent was obtained from all participants. The average completion time for the questionnaire was approximately 10–12 min.

3.5 Data Analysis

Quantitative data were analyzed using Python-based statistical methods, including descriptive and inferential techniques. Descriptive statistics were used to summarize key trends in AI tool usage among faculty members, including frequency distributions across demographic groups.

Chi-square tests were conducted to assess associations between categorical variables such as age group, gender, and institutional affiliation in relation to AI usage. Results revealed statistically significant relationships in some dimensions (e.g., age and AI engagement), while others (e.g., gender and region) showed no significant differences.

In addition to structured survey items, responses to open-ended questions regarding AI-related challenges were analyzed using thematic analysis. Thematic analysis was conducted following [16] six-phase approach, including familiarization with data, initial coding, generating themes, reviewing themes, defining and naming themes, and writing up the report. To strengthen the reliability of the qualitative analysis, coding was initially conducted by the primary researcher and reviewed by a second researcher to minimize potential bias (inter-rater reliability).

3.6 Limitations

Given the cross-sectional design and voluntary participation, findings may be influenced by self-selection bias and may not be fully generalizable to all regional higher education institutions. Additionally, while efforts were made to ensure the reliability of thematic analysis, full inter-rater consistency could not be guaranteed due to resource constraints.

4 Findings

4.1 Faculty Use of AI Tools in Teaching

Out of the 492 participants who responded to the AI-related survey items, the following patterns of usage were identified, see Table 1 and Fig. 1

Table 1. Faculty Use of AI Tools in Teaching

Usage Frequency	Percentage (%)
Occasional use (Bəzən)	24.4%
Frequent use (Tez-tez)	11.4%
Rare use (Nadir hallarda)	10.8%
Daily use (Hər gün)	3.8%
Never used (Heç istifadə etməyib)	17.0%

These findings suggest that while a considerable proportion of faculty members are aware of and engage with AI tools, regular and strategic integration into teaching practices remains low. This result is consistent with global studies emphasizing the growing presence of AI in academia but noting existing gaps in pedagogical implementation [5].

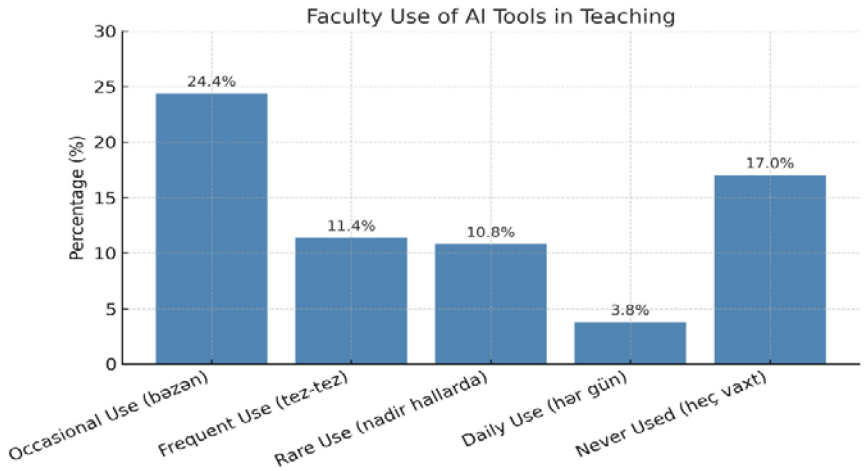


Fig. 1. Faculty Use of AI Tools in Teaching

4.2 Association Between AI Usage and Demographics

Chi-square tests were conducted to assess whether AI tool usage varied according to faculty demographics, see Table 2.

Age and AI Use: A statistically significant relationship was found ($\chi 2(24) = 55.07$, $p = 0.0003$). Younger faculty members (aged 26–37) reported significantly more frequent AI usage compared to older groups (50+). *Effect size (Cramér's $V = 0.21$) indicated a small to moderate association.*

Gender and AI Use: No significant association ($\chi 2(4) = 3.55$, $p = 0.47$). Male and female respondents reported similar levels of AI tool usage.

University Affiliation and AI Use: No significant differences across universities ($\chi 2(44) = 55.51$, $p = 0.114$).

These results highlight that age is the most prominent demographic predictor of AI engagement, while gender and regional institution affiliation do not significantly influence usage patterns.

4.3 Thematic Analysis of Open-Ended Responses

Respondents who used AI tools were asked to describe challenges they encountered. A thematic analysis was conducted based on 492 valid open-ended responses. Two researchers independently coded the data to enhance reliability. Three dominant themes emerged are shown in Table 3.

Cost and Access: Faculty members reported that high subscription fees and platform limitations restricted regular AI tool use.

Student Misuse: A major concern was the increasing reliance of students on AI tools (e.g., ChatGPT) to complete assignments, leading to risks of academic dishonesty and plagiarism.

Table 2. Association Between AI Usage and Demographics

Test result	Chi-square (χ^2)	p-value	Effect size (Cramér's V)	Interpretation
Age × AI Use	55.07 (df = 24)	.0003	0.21	Statistically significant. Younger faculty members use AI more frequently.
Gender × AI Use	3.55 (df = 4)	.47	–	No statistically significant difference.
University Affiliation × AI Use	55.51 (df = 44)	.114	–	No statistically significant difference.

Table 3. Dominant themes

Theme	Frequency
Cost and Access	267
Student Misuse	263
Lack of Training	113

Lack of Training: Respondents pointed out that there is minimal formal training or institutional support to guide responsible and effective AI integration.

These themes reinforce earlier studies by Perera [1, 14], stressing the need for improved ethical frameworks, infrastructure investment, and professional development to support AI adoption in higher education.

5 Discussion

The artificial Intelligence (AI) in HEIs is accepted as a transformative force globally, though its application remains inconsistence across various regions and contexts. This research is focused on faculty members' experiences in regional HEIs in Azerbaijan, a less explored subject in the global AI education discussions. The findings confirm several trends identified in international studies, as well as provide new insights into the challenges faced by educators in regional HEIs.

Findings show the growing awareness of AI tools among educators, yet limited readiness for its adoption into teaching processes. Frequent and planned use of AI tools still remaining low, approximately half of respondents reported using AI tools to some extent. This aligns with the research from [5], which confirms that AI awareness is rising while its pedagogical adoption is still in its early stages. Likewise, [1] discussed ad-hoc usage of AI tools like ChatGPT rather than systematic adopted use in education

processes. This highlights the need for a more systematic approach to AI adoption that goes beyond casual or isolated use.

The study also identified age as a significant factor in AI adoption, with younger faculty members showing a higher frequency of AI usage. This supports findings from [15], who discussed that younger digitally literate faculty are the more open to integrate new technologies. This suggests that the generational shift in teaching staff is likely to influence the future landscape of AI integration in higher education.

However, the research found no obvious variances in AI tools usage on gender or regional HEIs context, that being unexpected that other researchers confirmed gender-based differences in technology usage [7]. This situation may indicate two possible scenarios: regional HEIs provide equal levels of digital exposure to all educators, or AI readiness is equally low across genders and institutions.

A major theme in the open-ended responses was the concern over the limited access to high-quality AI tools, inadequate training, and ethical issues such as the misuse of AI by students. These findings align with ethical concerns raised in other studies, such as those by [6, 14], which emphasize the importance of regulating AI applications to avoid academic integrity issues, such as plagiarism and the production of superficial student work. These concerns suggest that, without proper training and ethical guidelines, the widespread use of AI could undermine the quality of education and research.

Institutional Support: One of the key shortcomings identified is the lack of institutional support mechanisms for AI adoption in regional universities. While some awareness exists, there is insufficient infrastructure, training, and access to AI tools that can enable faculty members to integrate these technologies into their teaching effectively.

Ethical Concerns: Faculty members expressed significant concerns about the ethical implications of AI usage, particularly regarding plagiarism and the potential for students to misuse AI tools [17–22]. This highlights a need for clear ethical guidelines and training on responsible AI use to ensure that AI enhances, rather than undermines, academic integrity.

Access to Tools: The limited access to high-quality AI tools remains a significant barrier. Many regional institutions lack the necessary resources to provide faculty with effective and affordable AI tools, hindering the potential for AI adoption at scale.

Training Deficiencies: A major gap in the adoption process is the lack of structured professional development programs focused on AI for faculty. Training and support are crucial for ensuring that AI is used effectively and responsibly in the classroom.

In summary, this research comes up with valuable findings and insights into the challenges and opportunities related to AI integration regional HEIs, particularly in Azerbaijan. The insights highlight the requirements for targeted institutional policies that cover the needs of regional HEIs including professional development, tools availability, and ethical policies. AI integration in regional HEIs will remain low with significant concerns related to academic integrity and technology misuse without such policies and frameworks being in place.

This research confirms the responsible AI adoption in HEIs is required to be inclusive, context-sensitive, and ethically grounded. At present regional HEIs face systemic

restrictions, though if they are provided the right tools, policies, frameworks, and professional development regional HEIs have the potential to lead digital innovation. As AI technologies continue to evolve, educational systems must also adapt, ensuring that no institution or educator is left behind.

5.1 Summary and Conclusion

This research investigated the AI tools awareness and usage among educators od educators in Azerbaijan regional HEIs. Conducting quantitative and qualitative analyses, the research comes up with insights, and comprehensive understanding of the situation with emerging AI technologies adoption by university faculty in under-resourced context.

The findings show that there is growing awareness, at the same time rare use of AI tools among relatively younger educators, though systematic and pedagogically planned use remains low. The absence of statistically significant differences across gender and institutional affiliation suggests that low adoption may be a shared experience rather than one specific to any demographic or regional group. However, the significant correlation between age and AI engagement highlights the importance of generational digital readiness in AI adoption.

Key barriers identified include limited access to AI platforms (due to cost or availability), misuse of AI by students (especially in academic assessments), and a lack of training and institutional support. These concerns echo global debates but are amplified in regional contexts where technological infrastructure and professional development opportunities may be scarce.

Importantly, this study offers a unique contribution by focusing on voices from regional universities- an often-overlooked group in global discussions of AI and education. By documenting their experiences, challenges, and perspectives, the study expands our understanding of how AI adoption unfolds beyond urban or elite academic settings.

In light of the findings, the following recommendations are proposed for policymakers, university leaders, and educators:

Invest in Faculty Training. Develop and deliver continuous professional development programs focused on AI literacy, ethical use, and practical classroom integration.

Ensure Equitable Access to AI Tools. Facilitate access to open-source or institutionally licensed AI platforms, reducing reliance on costly or commercial alternatives.

Promote Responsible AI Use Among Students. Incorporate academic integrity education related to AI tools into student orientation and course syllabi.

Establish Clear Institutional Guidelines. Develop policies that define acceptable AI usage in teaching and assessment, accompanied by support resources for faculty.

Support Ongoing Research and Evaluation. Encourage further studies to monitor the long-term impacts of AI on learning outcomes, teaching practices, and institutional equity.

This research affirms that responsible AI integration in higher education must be inclusive, context-sensitive, and ethically grounded. Regional universities, while currently facing systemic limitations, have the potential to lead innovation-provided they

are supported with the right tools, policies, and professional development. As AI technologies continue to evolve, so too must our educational systems, ensuring that no institution or educator is left behind.

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